

Student Guide — senior / module-09-subtraction-nikhilam

IKS Curriculum

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Student Workbook

Student Workbook — Subtraction — Nikhilam (All from 9) (Senior)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *Nikhilam Navataścaramam Daśataḥ* — _____
- *pratiyogi (modern coinage)* — _____

Day-by-day notes

Day 1 — The Sūtra — All from 9, Last from 10

What I learned:

One question I have:

Day 2 — Complement to 1000 and Beyond

What I learned:

One question I have:

Day 3 — Subtract from a Power of 10

What I learned:

One question I have:

Day 4 — General Subtraction via Complements

What I learned:

One question I have:

Day 5 — Why It Works — The Algebra

What I learned:

One question I have:

Day 6 — Bar Model Proof

What I learned:

One question I have:

Day 7 — Two's Complement in Computers

What I learned:

One question I have:

Day 8 — Speed Day

What I learned:

One question I have:

Day 9 — Mixed Practice

What I learned:

One question I have:

Day 10 — Final Assessment

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — The Sūtra — All from 9, Last from 10

Module: Subtraction — Nikhilaṃ (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute the complement of any 2-digit number to 100.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilaṃ Navataścaramaṃ Daśataḥ* (IAST: Nikhilaṃ Navataścaramaṃ Daśataḥ; lit. “all from 9, the last from 10”)

Primary source (today’s reading)

From *Bharati Krishna Tirthaji* — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Complement Flashcards* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces The Sūtra — All from 9, Last from 10.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See homework.md for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 2 — Complement to 1000 and Beyond

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute complements to 1000 and 10000 mentally.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *pratiyogi* (modern coinage) (IAST: pratiyogi (modern coinage); lit. “the number that adds to a base”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.

4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Complement to 1000 and Beyond.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 3 — Subtract from a Power of 10

Module: Subtraction — *Nikhilam* (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute 1000–567 and similar instantly.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *Nikhilaṁ Navataścaramaṁ Daśataḥ* (IAST: *Nikhilaṁ Navataścaramaṁ Daśataḥ*; lit. “all from 9, the last from 10”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Subtract from a Power of 10.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 4 — General Subtraction via Complements

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: use add-the-complement trick for arbitrary subtractions.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *pratiyogi* (*modern coinage*) (IAST: pratiyogi (modern coinage); lit. “the number that adds to a base”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces General Subtraction via Complements.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.

- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 5 — Why It Works — The Algebra

Module: Subtraction — Nikhilaṃ (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: derive the place-value identity that makes the rule work.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilaṃ Navataścaramaṃ Daśataḥ* (IAST: Nikhilaṃ Navataścaramaṃ Daśataḥ; lit. ““all from 9, the last from 10”“)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Why It Works — The Algebra.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 6 — Bar Model Proof

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: draw a visual proof using bar models and number lines.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *pratiyogi* (*modern coinage*) (IAST: *pratiyogi* (*modern coinage*); lit. “the number that adds to a base”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day's idea is — a number trick, a celestial pattern, a herb's identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today's idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Bar Model Proof.

Wrap-up (5 min)

- One-sentence exit ticket: “*What's one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 7 — Two's Complement in Computers

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: show how the same idea underlies binary computer subtraction.

Materials

- Whiteboard / chart paper
- Notebook, pen per student

- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilam Navataścaramam Daśataḥ* (IAST: Nikhilaṁ Navataścaramaṁ Daśataḥ; lit. “all from 9, the last from 10”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Two’s Complement in Computers.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 8 — Speed Day

Module: Subtraction — Nikhila (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: complete a 30-problem mental subtraction sprint.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *pratiyogi* (*modern coinage*) (IAST: pratiyogi (modern coinage); lit. “the number that adds to a base”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Speed Day.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 9 — Mixed Practice

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: solve a mixed problem set comparing complement vs traditional borrow.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilam Navataścaramaṁ Daśataḥ* (IAST: Nikhilam Navataścaramaṁ Daśataḥ; lit. “all from 9, the last from 10”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Mixed Practice.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 10 — Final Assessment

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: take the summative quiz.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *pratiyogi* (*modern coinage*) (IAST: *pratiyogi* (*modern coinage*); lit. “the number that adds to a base”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Final Assessment.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Writing Prompts

Writing Prompts — Subtraction — Nikhilam (All from 9) (Senior)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 300-word response.

1. In 3–4 sentences, explain Subtraction — Nikhilam (All from 9) to a friend who hasn’t taken this module.

2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Subtraction — Nikhilam (All from 9) with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Subtraction — Nikhilam (All from 9)? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?
9. Read one primary-source passage cited in this module. Translate or paraphrase it; comment on what it adds to your understanding.
10. Identify a contemporary debate about the IKS framework presented here (e.g., is Tirthaji's mathematics genuinely Vedic? are *doṣas* a valid framework today?). Take a position with two pieces of evidence.

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Homework

Homework — Subtraction — Nikhilam (All from 9) (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — The Sūtra — All from 9, Last from 10** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Complement to 1000 and Beyond** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Subtract from a Power of 10** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — General Subtraction via Complements** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Why It Works — The Algebra** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Bar Model Proof** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Two's Complement in Computers** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Speed Day** · Write 2 sentences about today's concept. (~10 min)

9. **Day 9 — Mixed Practice** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Final Assessment** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.